

When LSTMs Beat a Small Transformer on Chinese Name Generation

Abstract

This report revisits a Chinese name generation experiment under a stricter fairness-first protocol: fixed train/validation/test splits, three random seeds, parameter-matched neural models, structured metrics, and artifact-backed figures.

In this dataset and model-size regime, the parameter-matched LSTM achieved lower test perplexity than the small causal Transformer at 10%, 30%, and 100% training-data fractions. The result is not presented as a general claim that LSTMs are better than Transformers; it is evidence about a short, constrained micro-sequence task.

Headline result

At full data, the LSTM-small-matched model reached mean test perplexity 77.9209, compared with 86.7124 for the Transformer-small model across three seeds. The direction did not flip under the data-scale ablation.

The most important caveat is optimization: both models used the same flat Adam learning rate, with no Transformer-specific warmup or cosine schedule. Therefore, the result should be read as performance under this controlled shared training recipe, not as a final statement about the best possible Transformer recipe.

Task and Data

Why Chinese names are useful micro-sequences

Chinese full names are short, often only two to four characters, but they are not structurally empty. Generation depends on surname structure, local character compatibility, corpus bias, and stylistic regularities in historical data.

This makes the task a compact stress test for architectural inductive bias: the context window is tiny, but the output space remains culturally and distributionally constrained.

Dataset and split

The source data comes from the China Biographical Database (CBDB). The cleaned study dataset contains 584,442 full names. The split is fixed at 80% train, 10% validation, and 10% test with split seed 20260419. Training seeds are 13, 37, and 73.

Raw CBDB database files are treated as local-only data artifacts and are not redistributed in the public repository.

Vocabulary

Tokenization is a naive character-level mapping over individual Unicode characters observed in training names, plus special symbols such as SOS, EOS, PAD, and UNK. There is no BPE, unigram language-model tokenization, subword regularization, or compound-character tokenization.

The shared full-data vocabulary is about 8.4k tokens. A Chinese name is therefore represented as a short sequence of single-character vocabulary items rather than multi-character subword units.

Model and Training Configuration

The main comparison uses one parameter-matched pair. The LSTM is strictly forward/unidirectional. The Transformer is a decoder-only causal model with learned absolute positional embeddings, not RoPE or relative position encodings.

Item	LSTM-small-matched	Transformer-small
Parameters	5,469,500	5,355,708
Embedding dimension	160	256
Hidden / FFN dimension	hidden state 320	feed-forward 512
Layers	2 LSTM layers	2 Transformer encoder-style causal blocks
Directionality / causality	strictly forward, not bidirectional	causal self-attention mask
Attention heads	not applicable	8
Position encoding	recurrent state	learned absolute position embeddings, max_seq
Dropout	0.4	0.4
Optimizer	Adam	Adam
Learning rate	0.001 flat	0.001 flat
LR schedule	none	none: no warmup, cosine decay, or Transformer-
Max epochs / patience	40 / 5	40 / 5
Checkpoint selection	best validation perplexity	best validation perplexity

Optimization caveat: using the same flat Adam schedule is fair in the sense of controlling the training recipe, but it may disadvantage the Transformer. A future robustness check should include warmup plus cosine decay or another Transformer-tuned schedule.

Metric Definitions

Teacher-forced metrics and sample-based generation metrics are reported separately. Generation metrics use 2,000 unprompted samples per model/seed/temperature unless otherwise stated.

Metric	Definition
Perplexity	$\exp(\text{mean cross-entropy over non-PAD target tokens})$; lower is better
unique_name_ratio	Unique generated full names divided by total generated names.
Distinct-1	Unique generated characters divided by total generated characters
Distinct-2	Unique generated character bigrams divided by total generated characters
novelty_ratio	Generated names not appearing exactly in the training set divided by total generated names
train_overlap_ratio	Generated names appearing exactly in the training set divided by total generated names
duplicate ratios	Rates of consecutive or any repeated characters within generated names
invalid_output_ratio	Share of empty, malformed, or otherwise invalid generated outputs

Decoding uses pure temperature-scaled categorical sampling. No top-k truncation and no nucleus/top-p sampling are applied in these reported metrics. The main generation temperature is 0.8.

Stage 1: Full-Data Matched Pair

The full-data comparison favors the LSTM across all three seeds. The worst LSTM run remains below the best Transformer run, so the result is not driven by a single lucky seed.

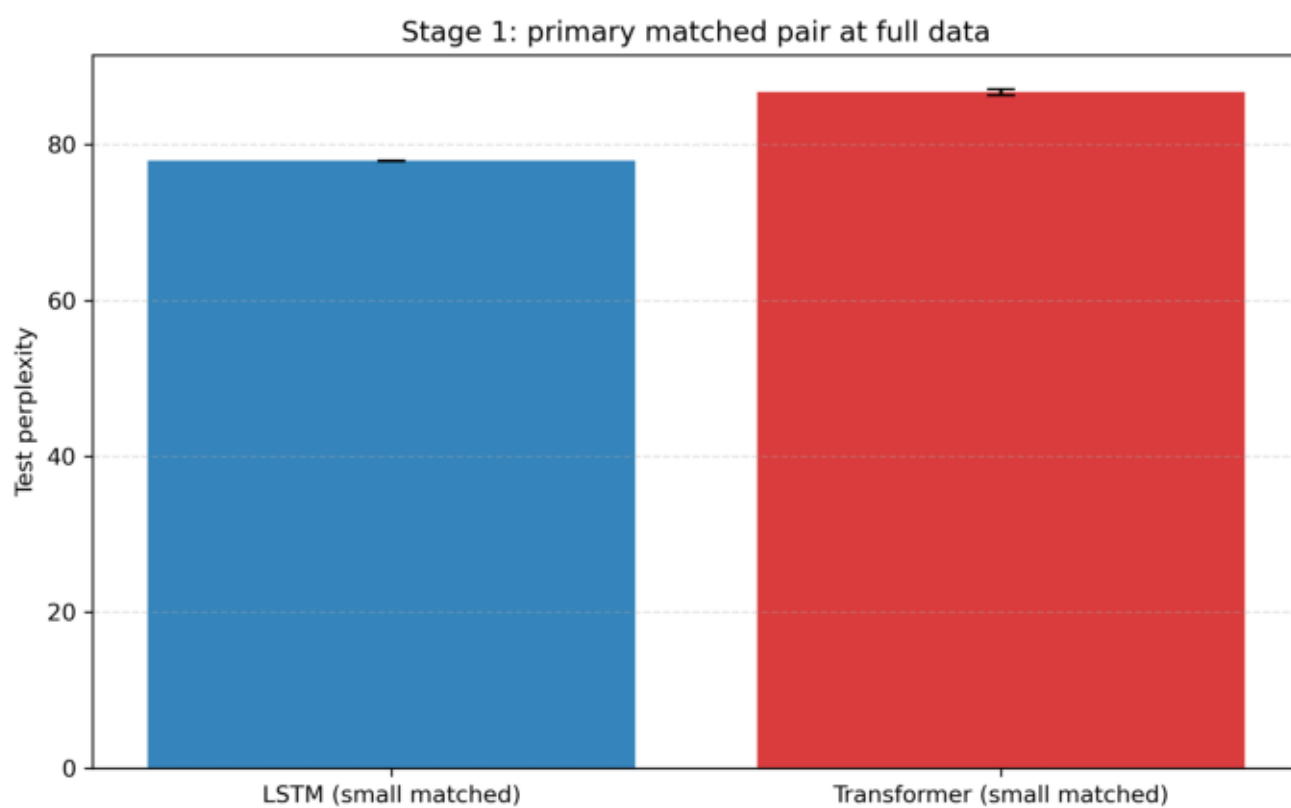


Figure 1. Mean test perplexity across three seeds for the primary matched pair at full data. Lower is better. Error bars represent seed-to-seed variation.

Convergence Behavior

The convergence curves reduce one simple concern: the Transformer was not obviously stopped before it had time to improve. However, they do not remove the optimization caveat, because a Transformer-specific schedule was not tested.

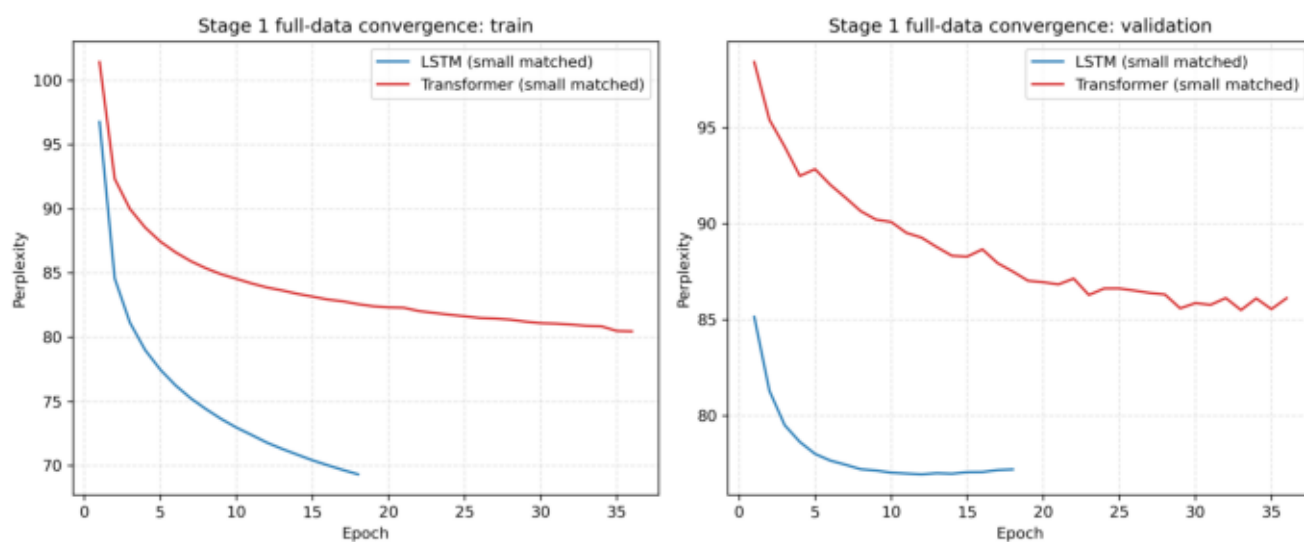


Figure 2. Mean train and validation perplexity across seeds. The LSTM reaches its best validation perplexity earlier, while the Transformer trains longer but plateaus higher under the shared flat-Adam recipe.

Distribution and Constraint Metrics

The result is not only about likelihood fit. Distribution-level metrics show differences in diversity, novelty, and train-set overlap, which helps separate memorization from useful variation.

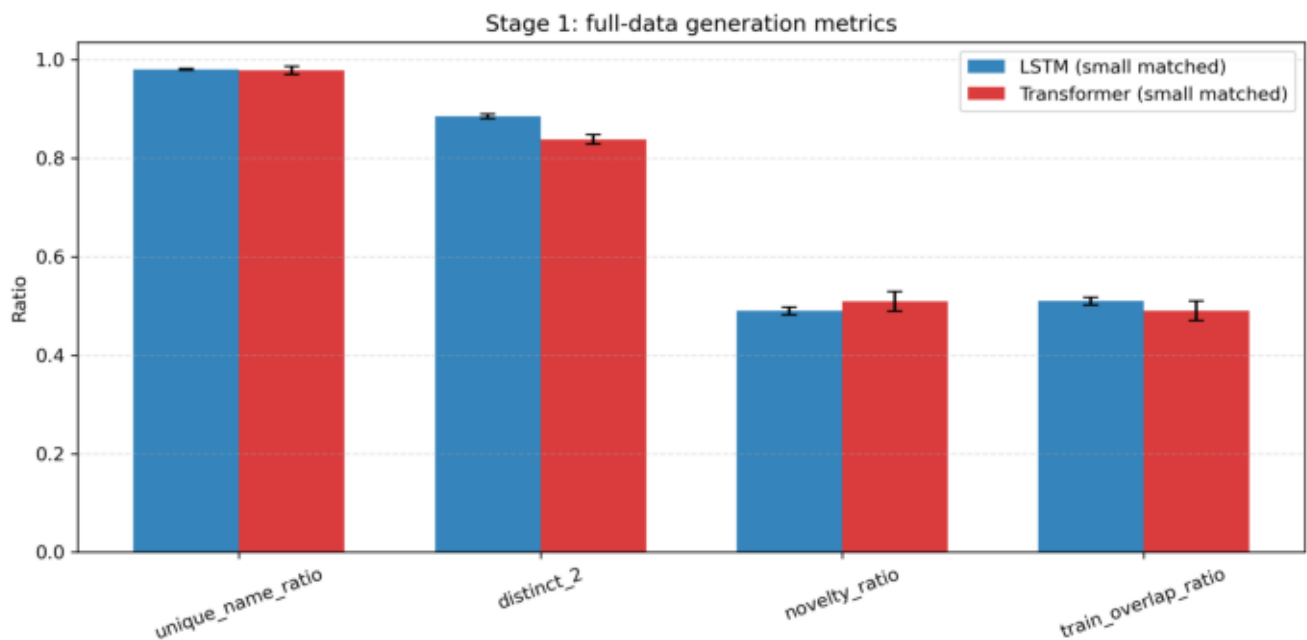


Figure 3. Full-data generation metrics at temperature 0.8. The LSTM has stronger character-bigram diversity, while the Transformer is slightly more novelty-seeking at the same temperature.

Constraint-Oriented Checks

Constraint metrics are included to avoid over-reading perplexity. For short name generation, local validity matters: duplicate characters, invalid outputs, and compound-surname compatibility are part of the product-facing quality story.

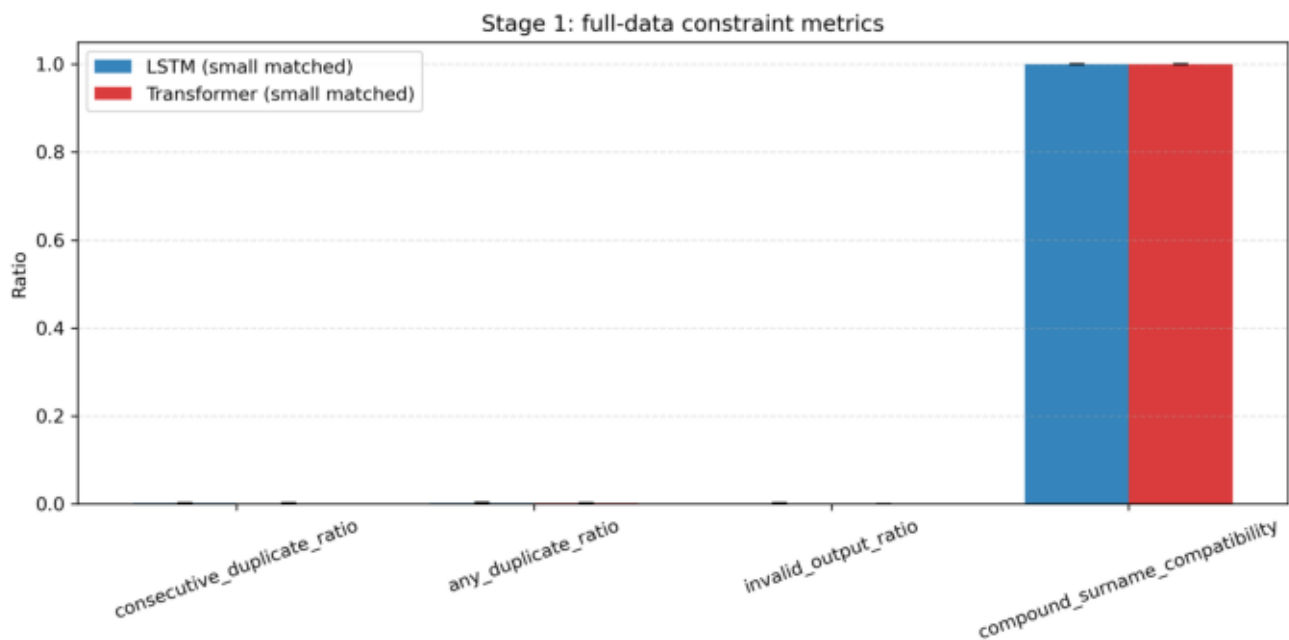


Figure 4. Duplicate and invalid-output rates remain low for both models. The headline result is not explained by one model simply producing broken strings.

Stage 2: Data-Scale Ablation

The same matched pair was retrained at 10%, 30%, and 100% of the training data. The LSTM advantage persisted at every data fraction.

Training fraction	LSTM test perplexity	Transformer test perplexity	LSTM advantage
10%	93.4379 +/- 0.1915	101.6190 +/- 0.1574	8.18
30%	84.7673 +/- 0.1920	96.9825 +/- 0.7187	12.22
100%	77.9209 +/- 0.0342	86.7124 +/- 0.4884	8.79

Values are mean +/- sample standard deviation over seeds 13, 37, and 73. Advantage is Transformer mean minus LSTM mean, so positive values favor the LSTM.

Data Scaling: Fit

This ablation makes the conclusion stronger than a single full-data run: the architecture gap is visible at 10%, 30%, and 100% training data.

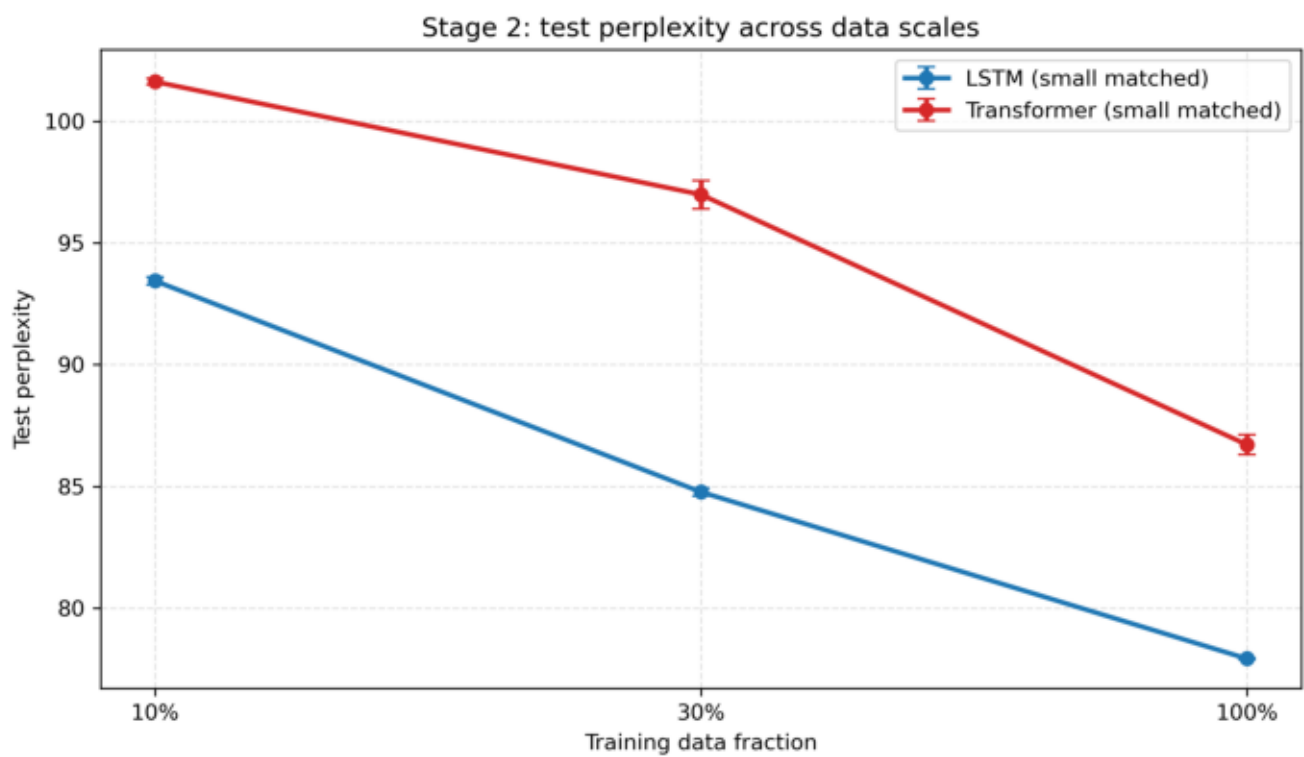


Figure 5. Test perplexity across data fractions. The direction of the result does not reverse when training data is reduced.

Data Scaling: Generation Behavior

The data-scale result is not one-dimensional. Fit, diversity, novelty, and memorization move differently as data increases.

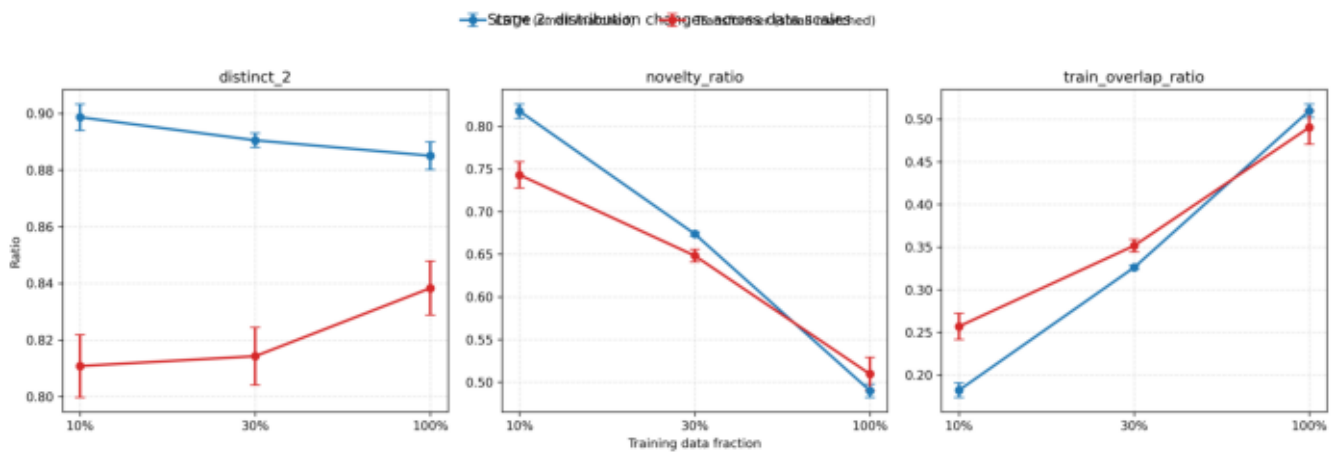


Figure 6. Distribution metrics across data fractions. At smaller data fractions, the LSTM is also stronger on novelty and train-overlap behavior; at full data, the Transformer becomes slightly more novelty-seeking while still worse on perplexity.

Interpretation, Product Extension, and Limits

Interpretation

The most defensible reading is narrow: under this controlled setup, dataset, model scale, and shared optimization recipe, the LSTM performed better on this short constrained sequence task. This is not evidence that LSTMs are generally superior to Transformers.

A plausible explanation is that a compact recurrent state is an efficient inductive bias for very short left-to-right generation. Another plausible explanation is that the small Transformer needs a better schedule, more capacity, or a different scaling regime.

Product extension

The web app separates fixed-token naming into two modes. Historical Pattern mode uses constrained search and reports corpus support, so it does not claim that a low-support character was impossible historically. Creative Ancient-Style mode is framed as constrained creative generation rather than historical reconstruction.

Current limits

The main claim comes from one matched size tier. The larger-capacity robustness pair was not completed. Human evaluation is not included in this report. The raw CBDB database is not redistributed. Finally, because no Transformer-specific learning-rate schedule was tested, optimization remains the most important open confounder.

A strong next step would rerun the Transformer with warmup plus cosine decay, then report whether the LSTM advantage remains under a Transformer-tuned recipe.